Benchmarking AI Models Efficiently

Siomara van Straalen*

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Abstract

Benchmarks that score several AI systems against one another with language-model judges, in the tradition of DRACO [Zhong et al., 2026], multi-agent debate [Du et al., 2024, Liang et al., 2024] and judge-based evaluation [Zheng et al., 2023], are rigorous and expensive. A faithful run multiplies task count by system count by judge count by repeated-scoring count by regeneration count by configuration-setting count. An interview study of nineteen practitioners describes evaluation practice "spanning informal 'vibe checks' to organizational meta-work" [van der Maden et al., 2026]; we take the cost of the rigorous end of that range to be the main reason such comparisons are run rarely or once, while the systems being compared keep changing. This paper writes the cost of a DRACO-style benchmark as an explicit function of its design parameters and derives six simplifications, each with a closed-form reduction factor and a statement of the guarantee it gives up. One result runs against intuition: judge count is usually the wrong place to economise. The combined reduction is a weighted harmonic mean of a generation-side and a judging-side factor, a ratio of total costs; the naive product of per-component factors overstates the saving by more than an order of magnitude. In the worked instance the combined reduction is roughly two orders of magnitude at that instance's setting-cost ratio and materially less at the ratios we measured. The appendices instantiate the algebra on published runs, on current commercial terms and on measurements made for this paper. A closing protocol runs the full benchmark once and each simplification beside it, so that the fidelity cost of every simplification is measured against its predicted saving. The author is not affiliated with the authors of DRACO; this is independent work that builds on their published benchmark.

*Correspondence through the paper's repository, https://github.com/SiomaraDAILFT/AI-Benchmarking. Text of this paper: © 2026 Siomara van Straalen, licensed under CC BY-NC 4.0 (free to share and adapt with attribution for non-commercial use; commercial use requires a licence from the author). Accompanying code: PolyForm Noncommercial License 1.0.0.

Contents

1 Introduction **3**

2 DRACO and Related Multi-System Evaluation Approaches **4**
2.1 DRACO . . . 4
2.2 Related and Competing Multi-System Approaches . . . 4

3 Merits and Costs of the DRACO Protocol **6**

4 A General Cost Model **7**

5 Six Cost-Reducing Simplifications **9**
5.1 S1: Reducing the configuration-setting sweep . . . 9
5.2 S2: Reducing answer regenerations . . . 10
5.3 S3: Reducing judge repeat-scorings . . . 10
5.4 S4: Batching judging across systems . . . 11
5.5 S5: Judge count, and the case against reducing it . . . 12
5.6 S6: Reducing task count, bounded by statistical power . . . 13

6 Cumulative Effect **14**

7 Statistical Constraints on the Simplifications **15**
7.1 Test, Aggregation, and Multiple Comparisons . . . 15
7.2 Power and the S6 Floor . . . 15
7.3 Blind and Calibrated Scoring . . . 16
7.4 Data Retention . . . 16

8 Empirical Validation of the Reduction Factors **17**

9 Limitations and Open Questions **18**

10 Conclusion **18**

A Published DRACO-Style Runs **19**
A.1 The benchmark . . . 19
A.2 Published scores . . . 19
A.3 Third-party trackers . . . 19

B A Worked Cost Instance at Current Commercial Terms **20**
B.1 Prompt-caching terms . . . 20
B.2 The setting-cost ratio α . . . 21
B.3 Call counts for a modest configuration . . . 21
B.4 The batching factor β . . . 21
B.5 The combined reduction . . . 21
B.6 The task floor n^* . . . 21

C Measurements Made for This Paper **23**
C.1 Judging-side premises (experiment E2) . . . 23
C.2 The setting-cost ratio α (experiment E1) . . . 23
C.3 Full method for experiment E1 . . . 24
C.4 Why the two measurements of α differ . . . 26
C.5 Competition problems as a task-family test (experiment E3) . . . 28

1 Introduction

A benchmark that compares several AI systems or configurations (individual models, ensembles, retrieval-augmented pipelines, multi-agent panels) against one another on a shared task set, scored by one or more language-model judges against a written rubric, is now a standard methodology. It underlies broad multi-scenario evaluations [Liang et al., 2023], crowdsourced pairwise-preference leaderboards [Chiang et al., 2024] and the automatic, judge-scored benchmarks built from them or checked against them [Li et al., 2024b, Dubois et al., 2024]. DRACO [Zhong et al., 2026] is a recent, well-specified instance and the running example here. We are not affiliated with DRACO's authors, and they have not reviewed this work. The literature on language models as judges, surveyed by Li et al. [2024a] and Li et al. [2025], supplies what each simplification costs; the subject here is the cost of the benchmark as a whole. The methodology is powerful for the same reasons that it is expensive to run faithfully. Task sourcing from real usage, several judges, repeated scoring to separate judge noise from differences between systems, and enough tasks to reach statistical power each add expense and running time, and each of them matters to the result.

That expense is the obstacle addressed here. van der Maden et al. [2026] found practitioners' methods ranging from informal "vibe checks" to organisational meta-work, with a gap between gathering evaluation data and acting on it; whether cost is what keeps teams from the rigorous end of that range is not something their study, or any other we know of, measures. We state it as a premise. Money is also only part of the expense. A rigorous multi-system evaluation takes time to set up: tasks and rubrics have to be written, several providers' interfaces connected, the judging blinded and randomised, every raw score retained and the statistics computed. The algebra below addresses the first cost; for the second, the tooling that runs the protocol of Sections 4–7 is published with this paper.[1] The set of systems worth comparing does not stay fixed. New models appear, existing systems are reconfigured, and a comparison that was current one quarter is stale the next. A methodology that can be afforded once a year cannot support decisions that are made continually. The question, then, is which parts of a rigorous protocol can be reduced, by how much, and at what cost to what the result can claim.

Section 2 describes DRACO and its neighbours, and Section 3 what DRACO does well and what it costs. Sections 4 to 6 give the cost model, the six simplifications and their combined effect; Section 7 the statistical constraints that bound them; Section 8 a protocol for checking predicted savings against measured fidelity loss; Section 9 the open questions. The appendices give the concrete figures: published DRACO-style runs (Appendix A), a worked instance at current commercial terms (Appendix B) and the measurements made for this paper (Appendix C).

[1] `https://github.com/SiomaraDAILFT/AI-Benchmarking`: the paper, the protocol runner and the measurement scripts.

2 DRACO and Related Multi-System Evaluation Approaches

2.1 DRACO

DRACO (Deep Research Accuracy, Completeness, and Objectivity) is a benchmark for deep-research systems published in 2026 by the developer of one such commercial system [Zhong et al., 2026, Perplexity AI, 2026a]. Its tasks are drawn from anonymised real usage of that system: de-identified user requests, first pre-filtered to a high-difficulty stratum, then filtered and augmented into a task set of 100 items spanning ten domains that the authors describe as "anonymized, open-ended and complex, objectively evaluable, and representative". Each task carries an expert-written scoring rubric of several tens of criteria, graded on four dimensions (factual accuracy, breadth and depth of analysis, presentation quality, and citation quality) by a language-model judge, with the grading repeated and averaged. The benchmark paper gives no sample-size or statistical-power guidance; the only public precision anchor we could find is a third-party replication that reports a standard error on its headline difference [Anand, 2026]. Appendix A records the published figures.

Selecting for "objective evaluability" is itself a form of selecting for gradeability, so the published task set is a curated sample from a filtered stratum rather than an unfiltered draw from real usage. The design still supports a weaker claim: each question began as a real user's request made for a real purpose, not as something a benchmark author wrote to be easy to grade, although the authors edited and augmented it before it entered the set.

2.2 Related and Competing Multi-System Approaches

Four structurally different kinds of system have reported DRACO scores or positioned themselves against it (Appendix A lists the published instances with their figures, judges and dates). First, *synthesis ensembles*: several models answer independently and a further model synthesises the answers; one API aggregator reports such ensembles, including a panel of inexpensive models, outscoring each constituent model alone [OpenRouter, 2026]. Second, the *benchmark author's own product*, whose scores should be read as vendor figures and not as independent confirmation [Perplexity AI, 2026b]. Third, a *multi-model research assistant* from a large software vendor, whose published replication reports a statistically significant improvement over a single-model baseline under a different judge model from the aggregator's [Anand, 2026]. Fourth, *retrieval-centred* deep-research products that report the highest published scores, describe multi-step retrieval across specialised sources, and do not describe multi-model ensembling [Valyu, 2026].

Two further categories are sometimes conflated with these. A *router* [Not Diamond, 2024, Martian, 2023] selects a single best model per query; it does not combine multiple models' outputs and answers a different question from a multi-system panel. *Consumer parallel-chat products* [AllChat, 2026], which send one question to several assistants and return a consensus answer, are structurally closer to a panel but do not, to our knowledge, publish a benchmark score of any kind.

Because judge choice alone moves published DRACO scores by a wide margin (Appendix A), comparisons across those numbers should be read as directional at best. For that reason S5 (Section 5) argues against reducing the number of judges, and Section 9 explains that the interval of Section 7 is conditional on the judges used.

Combining several models' outputs has its own evidence base: sampling several answers from one model and keeping the most consistent one [Wang et al., 2023b], ranking and fusing the candidates of several open models [Jiang et al., 2023], and layered mixture-of-agents designs, one of which, built only from open models, reached 65.1% on a judge-scored instruction-following benchmark against

57.5% for GPT-4 Omni [Wang et al., 2024]. That benchmark scores length-controlled win rates because its judge favours longer answers [Dubois et al., 2024], which is one reason Section 7 treats the judge as a source of error. Multi-agent debate over several rounds improves factual accuracy and reasoning relative to a single instance [Du et al., 2024, Liang et al., 2024]. The same literature supplies a caution. Zhang et al. [2025] argue that reported debate gains are often confounded with extra compute or more diverse models, and Cemri et al. [2025], who annotated more than 1,600 multi-agent traces, sort the failures into three groups: system design, misalignment between agents, and task verification. Both are reasons to take the statistics of the scoring (Section 7) as seriously as the comparison itself.

3 Merits and Costs of the DRACO Protocol

Drawing tasks from real usage rather than composing them for the benchmark addresses a documented weakness of curated benchmarks: their construct validity, the degree to which they measure what they are framed as measuring, is often unexamined [Raji et al., 2021], and dataset design and annotation reliability bound what a benchmark can show [Bowman and Dahl, 2021]; sourcing from real users is now an established design for that reason [Chiang et al., 2024, Lin et al., 2024]. A multi-dimension rubric graded per task is more informative than a pass/fail judgement or a generic quality prompt: Ye et al. [2024] show that "the fine-graininess of evaluation is crucial for attaining a holistic view of model performance and increasing the reliability of the evaluation", and rubric-conditioned judging against user-defined criteria has become the standard design of purpose-built evaluator models [Kim et al., 2024]. Repeated judge scoring separates judge noise from signal; S3 in Section 5 gives the evidence that the noise is real.

Against that, no sample-size or statistical-power methodology has been published for DRACO. Judge choice moves absolute score levels: the benchmark's own authors report that one of their three judges "consistently assigned lower scores" [Zhong et al., 2026, Appendix A], and the effect is general [Thakur et al., 2025]. Third-party trackers of DRACO results already disagree about which system leads and by how much (Appendix A); no figure should yet be taken from them.

The weakness that matters most for whether the methodology gets used at all is a third one: run rigorously, DRACO or any protocol with its structure is expensive and slow. A single run multiplies how many systems are compared, how many tasks, how many judges, how many times each judge re-scores an answer, how many times each answer is regenerated, and how many configuration variants (temperature, retrieval settings, reasoning-effort levels) are swept per system. Each factor is well motivated on its own, and Section 4 makes each explicit; their product is what makes a faithful run too costly to repeat.

4 A General Cost Model

We define the following parameters, each a property of the benchmark's own design and independent of any particular system under test:

- C: the number of systems or configurations under comparison.
- T: the number of tasks.
- E: the number of configuration-setting variants swept per system (for example, a minimum and a maximum reasoning-effort or compute-budget setting).
- R: the number of independent answer regenerations per (system, task, setting) triple, used to separate a system's true output distribution from a single unlucky or lucky draw.
- J: the number of independent judges.
- K: the number of repeated scorings per (judge, answer) pair, used to separate judge test-retest noise from a real difference.
- g: the average cost of one generation call, that is, of obtaining one answer from one system, for one task, at one setting. It is treated as an observable, external quantity (a published API price, or a metered cost the system itself reports) and is not decomposed further; the framework is agnostic to how any system produces its answer.
- j: the average cost of one judging call, that is, of one judge scoring one answer once, unbatched.

Generation cost. Every system, at every setting, is queried R times per task:

$$G = C \cdot E \cdot R \cdot T \cdot g. \tag{1}$$

Judging cost, unbatched. Every one of the $C \cdot E \cdot R \cdot T$ generated answers is scored by every judge, K times each:

$$J_u = C \cdot E \cdot R \cdot T \cdot J \cdot K \cdot j. \tag{2}$$

Judging cost, batched. If a judge is instead shown all C systems' answers to the same (task, setting, regeneration) triple in a single call and asked to score each, the C factor drops out of the call count, at the cost of a larger per-call price j':

$$J_b = E \cdot R \cdot T \cdot J \cdot K \cdot j'. \tag{3}$$

Decomposing j and j' into a fixed component (task description, rubric text, formatting instructions, paid once per call regardless of how many answers it covers) and a marginal component (the cost of reading and scoring one additional answer), $j = j_{\text{fixed}} + j_{\text{marginal}}$ and $j' \approx j_{\text{fixed}} + C \cdot j_{\text{marginal}}$, so batching saves

$$J_u - J_b = E \cdot R \cdot T \cdot J \cdot K \cdot (C - 1) \cdot j_{\text{fixed}}. \tag{4}$$

The saving is largest when the fixed overhead (rubric, task description, instructions) is large relative to the marginal cost of reading one more answer. That ratio is an empirically checkable property of a given rubric and task format and does not depend on any system.

Effect of prompt caching. Equation 4 assumes the fixed component is paid in full on every unbatched call. The largest commercial API providers now bill a repeated prompt *prefix* at a discounted cached-read rate, in some cases with a premium on the first, cache-writing call, a minimum prefix length and a limited cache lifetime; output tokens are not cached [OpenAI, 2026a, Anthropic, 2026a, Google, 2026]. Appendix B records the terms in force at the time of writing. A task description together with a rubric of several tens of criteria will usually exceed the published minimum prefix lengths, although not always: a forty-criterion rubric on its own measured below

one provider's minimum and drew no cached tokens (Appendix C), so the prefix length should be checked. The C calls that score one task's answers naturally fall within a single cache lifetime. Writing c for the cached-read price as a fraction of the standard input price and w for the price of the first, cache-writing call as a multiple of the standard input price ($w = 1$ where no write premium is charged), the unbatched judging cost with caching is

$$J_u^{\text{cached}} = E \cdot R \cdot T \cdot J \cdot K \cdot \left[\left(w + (C-1)\, c\right) j_{\text{fixed}} + C \cdot j_{\text{marginal}} \right], \tag{5}$$

which reduces to Equation 2 at $c = w = 1$. We are not aware of a published treatment of prompt caching's effect on the cost of judged evaluation; batch prompting has been studied as a cost measure [Cheng et al., 2023] and batch-wise evaluation as a quality measure [Yuan et al., 2024], but caching enters only through provider pricing pages. Section 5 (S4) shows that caching weakens the case for batching considerably.

Total cost, using batched judging:

$$\text{Total} = G + J_b = E \cdot R \cdot T \cdot \left(C \cdot g + J \cdot K \cdot j'\right). \tag{6}$$

Equation 6 shows the problem: three of the six design choices (E, R and T) multiply the whole cost, and each of the other three multiplies one of its two terms. The number of generation calls is $C \cdot E \cdot R \cdot T$ and the number of unbatched judging calls is that quantity multiplied again by $J \cdot K$, so a configuration that looks modest on every individual dimension implies call counts in the thousands and tens of thousands respectively before any per-call price is attached to them (Appendix B evaluates one such configuration).

5 Six Cost-Reducing Simplifications

Each simplification below states its reduction factor in closed form and, separately, what it trades away. Nothing here recommends adopting all of them.

5.1 S1: Reducing the configuration-setting sweep

If E settings are swept per system and one setting's generation cost is α times a cheaper baseline setting's cost (for example, an extended-reasoning or higher-compute setting against a minimal one), reducing to the single cheapest setting saves a factor of

$$\left.\frac{\sum_{e=1}^{E} g_e}{g_{\min}}\right|_{E=2} = 1 + \alpha \tag{7}$$

on generation cost, and a separate factor of E on judging cost (fewer conditions to judge). Equation 7 assumes the *cheap* setting is the one retained. If instead the expensive setting is retained, which for reasons given below is often the appropriate choice, the generation-side factor is $(1 + \alpha)/\alpha$, which tends to 1 as α grows; S1 then saves almost nothing on generation and only the factor E on judging. If the two settings cost the same ($\alpha = 1$), either choice saves the trivial factor of 2. The generation-side case for S1 rests on α being large and on the cheap setting being the operating point of interest. Two pieces of evidence bear on α. The largest providers bill a reasoning model's internal "thinking" tokens as output tokens and expose the amount as a per-request setting, so a call at a high setting pays the same rate on a much larger token count [OpenAI, 2026b, Anthropic, 2026c]. An independent measurement service that runs a fixed suite at each setting reports, for one frontier model, more than an order of magnitude more output tokens at the highest setting than at the lowest, with a substantial difference in measured quality [Artificial Analysis, 2025]; Appendix B gives the figures, and the same service publishes per-task token usage for every model it evaluates [Artificial Analysis, 2026], so α can be looked up. It should also be measured. On open-ended analytic prompts we found α between 1.75 and 3.2 for two frontier models (Appendix C), an order of magnitude below that published figure; α depends on the model, on how its effort setting is implemented and on the task family.

What is lost. Questions about the contrast between settings can no longer be answered, since only one setting is measured. A less obvious loss matters more. Measuring every system at its cheapest setting and reporting a ranking of the *systems* assumes that the order at the cheap setting predicts the order at the expensive one, and the evidence runs the other way. Guedes de Souza and Panisson [2026] vary the generation budget, the maximum number of output tokens, over seven levels from 64 to 4,096 for four models on three reasoning benchmarks and find that "model rankings reverse across budgets on all benchmarks" ($p < 0.01$), with 3–19% of items losing accuracy as the budget grows. The effect of extra compute is itself uneven. It can let a smaller model beat one 14× larger at equal compute, on problems the smaller model already solves some of the time [Snell et al., 2024], or one five times larger [Wu et al., 2025], lift an open code model from 15.9% to 56% of tasks solved with 250 samples [Brown et al., 2024], and at the same time lower accuracy on hard queries while raising it on easy ones [Chen et al., 2024]; in one automated-scoring task higher effort helped on average but its benefit "varied by model family", and the most accurate configuration ran at low effort [Frohn, 2026]; and on closed-book knowledge questions, answered without search, it "does not consistently improve accuracy and often leads to more hallucinations" across 14 reasoning models, although for GPT-5-mini with a search tool, closer to DRACO's setting, it raised accuracy from 41.2% to 53.0% and cut hallucinations from 16.6% to 8.0% [Zhao et al., 2026a]. Kim and Jung

[2026] accordingly ask that the effort setting, its rationale and whether it was left at the vendor default be reported whenever several models are benchmarked.

The consequence for S1 is a scope condition. A comparison run at one setting answers the question "which system is best *at this operating point*", and its result should be reported with the setting named, as Guedes de Souza and Panisson [2026] recommend; it does not license a claim about the same systems at any other setting. Where the expensive setting is the one that would be used in practice, the appropriate reduced design measures at *that* setting alone and accepts the smaller generation-side saving of $(1+\alpha)/\alpha$. S1 removes the sweep; it does not change the operating point. Where both settings matter, the pilot of Section 8 should include a small two-setting subset and report the rank agreement between settings (Kendall's τ, or the sign of each pre-registered difference) before the sweep is dropped.

5.2 S2: Reducing answer regenerations

Reducing R regenerations to 1 saves a factor of R on both generation cost (Equation 1) and judging cost (Equations 2–5), since both scale linearly with R. **What is lost**: any way of telling whether a single answer was a representative draw from the system's output distribution or an unusually good or bad one. That distribution is not degenerate even at temperature zero. An open-weight model (Qwen3-235B), sampled 1,000 times at temperature zero on the same prompt, returned 80 distinct completions, a consequence of server-side batching acting through floating-point non-associativity [He and Thinking Machines Lab, 2025], and in an open-weight reasoning model the same numerical cause moves greedy-decoding accuracy by up to 9 points with batch size and hardware [Yuan et al., 2025]. At the level of evaluations, accuracy varied by up to 15 points across ten "deterministic" runs of five models on eight tasks [Atil et al., 2024]; between 48% and 76% of code-generation tasks produced no two identical outputs across repeated requests, with temperature zero reducing but not removing the effect [Ouyang et al., 2023]; the choice between greedy decoding and sampling shifts measured performance [Song et al., 2024]; and Madaan et al. [2024], measuring the variance across seeds and checkpoints, "encourage practitioners to carefully factor in variance when comparing models". At $R = 1$ this within-system variance is not removed; it goes unmeasured and is absorbed into whatever difference the benchmark reports. Where S2 is applied, the decoding configuration should be fixed and stated, and the pilot of Section 8 should include a small multi-regeneration subset so that the size of the ignored variance is at least estimated.

5.3 S3: Reducing judge repeat-scorings

Reducing K repeats to 1 saves a factor of K on judging cost only (Equations 2–5); generation cost is unaffected, since K only multiplies the scoring step. **What is lost**: test–retest reliability, that is, whether a single judge's score on a given answer is stable when the judge is asked again with no memory of the prior scoring. The concern is documented for language-model judges. Stureborg et al. [2024] report low agreement between repeated samples of the same evaluator on the same item, alongside sensitivity to prompt differences that a human reader would consider immaterial; Schroeder and Wood-Doughty [2024] treat judge reliability in the classical psychometric sense, estimate it with McDonald's ω over repeated ratings, and conclude that "a single sample from the model's probability distribution can still be misleading"; and Thakur et al. [2025], across 13 judge models, find that even the best-aligned judges' scores "may still differ with up to 5 points from human-assigned scores", with a general tendency towards leniency. DRACO's own published protocol averages over five grading runs per answer and reports the resulting standard deviation [Zhong et al., 2026], as does the third-party replication [Anand, 2026]. A judge whose repeated

scores on the same answer spread widely is indistinguishable, at $K = 1$, from a judge whose single score happened to be reliable; in a small measurement made for this paper, five scorings of one answer by one frontier judge spanned 17 points on a 100-point scale (Appendix C). Because judge repeats are usually inexpensive relative to generation (the argument of S5 below), and because the reliability they measure is a property of the judge and not of the systems under test, the economical course is to estimate it once, in the pilot, with $K > 1$ on a subset, and to set K for the main run from that estimate. One further result bears directly on how many repeats are needed. Where the judging interface exposes the probabilities the judge assigns to each candidate score token, a single call yields the *expected* score over that distribution, and Wang et al. [2025] find that "taking the mean of the judgment distribution consistently outperforms taking the mode (i.e. greedy decoding) in all evaluation settings", that is, pointwise, pairwise and listwise. Repeated sampling at non-zero temperature is a Monte Carlo approximation of that same mean, so where score probabilities are available, one call replaces the K repeats for the purpose of averaging out sampling noise; $K > 1$ then remains useful only for position calibration (Section 7.3) and where the interface offers no probabilities.

5.4 S4: Batching judging across systems

Showing a judge all C systems' answers to the same task in one call, instead of scoring each system's answer separately, reduces the judging term by the factor

$$\beta \equiv \frac{J_u^{\text{cached}}}{J_b} = \frac{w + (C-1)\,c + C\rho}{1 + C\rho}, \qquad \rho \equiv \frac{j_{\text{marginal}}}{j_{\text{fixed}}}, \tag{8}$$

with c and w the cached-read fraction and cache-write multiple of Equation 5. Without caching ($c = w = 1$) this is $C(1+\rho)/(1+C\rho)$: it approaches C as $\rho \to 0$ (the fixed overhead dominates the per-call cost, so batching amortises it across all C answers) and approaches 1 as $\rho \to \infty$ (the per-answer reading cost dominates, and there is nothing left to amortise). With caching and no write premium ($w = 1$), the corresponding limits are $1 + (C-1)\,c$ and 1: the batching factor can never exceed $1 + (C-1)\,c$, which at the cached-read fractions currently published (Appendix B) stays well below C and grows by only c for each system added (1.4 at $C = 5$ and $c = 0.1$, against 5 without caching). The cost case for S4 is therefore weak wherever prompt caching is available, which is now the default behaviour of several large providers (Section 4). **What is added rather than lost**: a judge shown several systems' answers to the same task side by side has calibration context it would lack when scoring one answer in isolation. That this helps has been measured: Yuan et al. [2024] find that scoring one sample at a time is "sensitive to prompt design" and shows "poor resistance to noise", and that presenting a batch of samples in one prompt, so that in-batch comparison serves as a second reference alongside the rubric, "outperforms state-of-the-art methods by 10.5% on Pearson correlations with only 64% API cost on average". The same direction appears in the comparative-assessment literature: pairwise comparison correlates better with human judgement than isolated absolute scoring [Liusie et al., 2024, Liu et al., 2024b], and isolated scoring is the format in which the best-known judge protocol, G-Eval, reaches only a Spearman correlation of 0.514 with human raters on summarisation [Liu et al., 2023]. Batching is a candidate methodological improvement (Section 7.3) as well as a cost measure.

Batching has costs of its own. The judge may score *relative* to the batch instead of independently against the absolute rubric, even when instructed to do the latter. The distinction matters, because purely comparative judgements are more easily moved by features irrelevant to quality: Tripathi et al. [2025] find that pairwise preferences flip in about 35% of cases when a distractor feature is

introduced, against 9% for absolute scores, and conclude that "absolute scoring is more robust to such manipulation". The protocol we recommend, an absolute rubric with the batch present only as reference, is the point between the two formats that the evidence favours. The judge can be held to it only by instruction, and instructions are imperfectly followed. A C-way listwise prompt is the format language models handle least reliably: Qin et al. [2024] argue that "off-the-shelf LLMs do not fully understand" pointwise and listwise ranking formulations, and Cheng et al. [2023] report that batched-prompt accuracy depends on "the number of samples in each batch and the complexity of tasks". And the longer multi-answer prompt exposes the judge to the documented degradation in how language models use information placed in the middle of a long context [Liu et al., 2024a], and to position bias, which Section 7.3 treats with its sources. Anonymised, order-randomised presentation with a small C reduces these costs without removing them; under caching the trade is made for calibration, since the cost saving is small.

5.5 S5: Judge count, and the case against reducing it

We recommend against this one. Fewer judges are cheaper, and usually not by much. From Equation 3, the absolute saving from reducing J judges to 1 is

$$\Delta_J = E \cdot R \cdot T \cdot (J-1) \cdot K \cdot j', \tag{9}$$

which is a fraction Δ_J/Total of the whole budget (Equation 6) equal to

$$\frac{(J-1) \cdot K \cdot j'}{C \cdot g + J \cdot K \cdot j'}. \tag{10}$$

Whenever generation cost dominates judging cost, that is, whenever $C \cdot g \gg J \cdot K \cdot j'$, this fraction is small. A cheaper judge is not enough on its own for that: $J \cdot K$ is often larger than C (15 against 5 in the worked instance of Appendix B), so a generation call has to cost several times a judging call. Reducing J saves little in absolute terms and removes the one thing judge count provides, evidence about whether the result depends on which judge was used. A single judge still yields a confidence interval across *tasks*; what it cannot yield is any estimate of judge-to-judge variation, so the resulting interval is conditional on that one judge and treats its idiosyncrasies as ground truth (Section 9 returns to this in the language of generalisability theory). Judge idiosyncrasy is well documented: the same answer receives different scores from different judge models, judges differ in leniency, and their absolute levels are not interchangeable [Thakur et al., 2025, Zheng et al., 2023]; DRACO's own authors found that one of their three judges "consistently assigned lower scores than the other two" [Zhong et al., 2026].

Judge count buys independent evidence only if the judges' errors are independent. Kohli [2026] shows that a panel of nine frontier models from seven families, tested on three natural-language-inference datasets, delivered only about two effective votes' worth of information, because roughly three quarters of the panel's nominal independence was lost to shared errors on the same items; the panel's accuracy fell 8–22 points short of what independent voting would have achieved, and the best single model matched or exceeded the whole panel. Nor is diversity of provider a sufficient remedy. Across more than 350 models, Kim et al. [2025] find that "larger and more accurate models have highly correlated errors, even with distinct architectures and providers"; on one leaderboard dataset "models agree 60% of the time when both models err"; and they trace the consequence into LLM-as-judge evaluation directly. Judge independence has to be measured: inter-judge agreement, or an effective-sample-size estimate of the kind Kohli [2026] uses, should be reported so that a reader can see how much independent evidence the panel contains, a reporting standard Yang et al.

[2026] also call for after showing that "an LLM-as-judge score can move even when the candidate responses stay fixed, simply because the evaluator has changed". On the aggregation side, methods that model the judges' shared confounder explicitly recover part of what correlation costs [Zhao et al., 2026b]; they presuppose that the per-judge scores were retained, which Section 7 requires in any case. More judges need not cost more, either. Verga et al. [2024] find that a panel of several smaller models from different families correlates better with human judgement than a single large judge, exhibits less intra-model bias, and costs over seven times less. Under Equation 3 the relevant quantity is the product $J \cdot j'$: a panel of three inexpensive judges can cost less than one expensive judge while providing the judge-generalisability evidence a single judge cannot.

Before any dimension is reduced, find the term of Equation 6 it controls and reduce the one that is expensive.

5.6 S6: Reducing task count, bounded by statistical power

Reducing T saves linearly on both generation and judging cost, but unlike S1–S5 it has a separate, explicit consequence: below a computable floor n^* (Section 7.2), the benchmark can no longer detect the effect size it was designed to detect at the stated confidence level. This is the one dimension where the cost-reduction objective and the statistical-validity objective trade off directly against each other, and its floor can be computed once a pilot has produced a variance estimate. **What is lost**: statistical power alone, provided the reduced task set is representative of the full one. That proviso is where the evidence from the benchmark-compression literature is relevant. Maia Polo et al. [2024] show that "to accurately estimate the performance of an LLM on MMLU ... consisting of 14K examples, it is sufficient to evaluate this LLM on 100 curated examples"; Perlitz et al. [2024a] report reductions in evaluation compute "by x100 or more" on HELM with "minimal loss of benchmark reliability", while warning that "a benchmark leader may change by merely removing a low-ranked model from the benchmark"; and Vivek et al. [2024] find that "1–30 anchor points outperforms uniform sampling and other baselines at accurately ranking models". Those figures do not transfer directly. The compressed benchmarks in that literature start from thousands of multiple-choice items with a large pool of previously evaluated models from which item informativeness can be estimated; a DRACO-style benchmark starts from about a hundred open-ended tasks (DRACO has 100) and no such pool, so the curated-subset methods are available only once several rounds of retained raw data (Section 7) exist. And any informed selection of tasks must itself be fixed before the systems are run, or task selection becomes one more researcher degree of freedom [Simmons et al., 2011]. Until then, S6 reduces T by pre-registered stratified sampling from the task domains, and Equation 13 sets the floor. The endpoint towards which S6 develops, once several rounds of retained raw data exist, is already described in that literature: adaptive item selection driven by an item response model, which Hofmann et al. [2025] show attains higher validity and lower variance on MMLU "with fifty times fewer items", and subset evaluation with confidence intervals of valid frequentist coverage, for which Wu et al. [2026] report effective-sample-size gains of up to five times over uniform sampling.

6 Cumulative Effect

The simplifications do not act on the same quantity. S1's factor $(1+\alpha)$ acts on generation cost; S1's factor E, S3's factor K and S4's factor β act on judging cost only; S2's factor R and S6's factor T_{full}/n^* act on both. A single product of all of them applied to the total would credit the whole budget with a saving that only its judging share can deliver. The exact combined reduction is a ratio of totals. Take the full protocol to judge each answer in its own call, as DRACO does, so that its judging cost is $J_{\text{full}} = J_u^{\text{cached}}$ (Equation 5; J_u where caching is unavailable), and let s be the share of the full protocol's cost that is judging, $s = J_{\text{full}}/(G_{\text{full}} + J_{\text{full}})$. Define the two component factors

$$F_G = (1+\alpha) \cdot R \cdot \frac{T_{\text{full}}}{n^*}, \qquad F_J = E \cdot R \cdot K \cdot \beta \cdot \frac{T_{\text{full}}}{n^*}, \tag{11}$$

with β from Equation 8 ($\beta = 1$ if S4 is not applied), and with the first factor of F_G replaced by $(1+\alpha)/\alpha$ if S1 retains the expensive setting instead of the cheap one. The reduced total is $G_{\text{full}}/F_G + J_{\text{full}}/F_J$, so

$$\text{Reduction} = \frac{G_{\text{full}} + J_{\text{full}}}{G_{\text{full}}/F_G + J_{\text{full}}/F_J} = \left(\frac{1-s}{F_G} + \frac{s}{F_J}\right)^{-1}, \tag{12}$$

a weighted harmonic mean of F_G and F_J, which always lies between them. As $s \to 0$ the reduction tends to F_G, so when generation dominates the budget, which by S5 needs a generation call to cost several times a judging call, the judging-side simplifications S3 and S4 contribute almost nothing to the combined figure, however large their own factors. The naive product $F_G \cdot E \cdot K \cdot \beta$ (equivalently, all factors applied to the whole budget) exceeds the exact reduction by a factor that grows with $E \cdot K \cdot \beta$ and exceeds an order of magnitude for typical values (Appendix B gives the figures). And because F_G and F_J share the factors R and T_{full}/n^*, S2 and S6 are the only simplifications whose benefit is independent of s; they are also the two whose cost to validity is measurable in the pilot (Sections 5 and 7.2).

Appendix B evaluates Equation 12 for an illustrative configuration: roughly two orders of magnitude, with and without prompt caching, across plausible values of s, at that instance's setting-cost ratio α. At the values of α we measured on analytic prompts (Appendix C) the same configuration yields well under a hundredfold, because α enters F_G directly.

7 Statistical Constraints on the Simplifications

7.1 Test, Aggregation, and Multiple Comparisons

The quantity of interest for each pre-registered comparison between two systems is the task-level paired difference in score. A paired design is used throughout because every system answers the identical task set, and task difficulty is shared across systems: as Miller [2024] put it, "because eval question scores are likely to be positively correlated, even across unrelated models, paired differences represent a 'free' reduction in estimator variance when comparing two models". The test is *one-sided* when the research question is directional ("does system A outperform system B"), and *pre-specified before any data are collected*: a paired permutation (sign-flip) test on the task-level differences, which is exact and distribution-free under the null hypothesis of exchangeable signs [Ernst, 2004], and which does not rely on the central-limit approximation that Bowyer et al. [2025] show to understate uncertainty for evaluations with "fewer than a few hundred datapoints", which is the regime every task count in this paper falls into. We recommend *against* testing the differences for normality and then choosing between a t-test and a non-parametric alternative. The simulation evidence on whether that two-stage procedure inflates the Type I error rate is mixed [Schucany and Ng, 2006, Rasch et al., 2011, Rochon et al., 2012], and the recommendation does not depend on it: a branch chosen after seeing the data is a researcher degree of freedom of the kind that "dramatically increases actual false-positive rates" [Simmons et al., 2011], and a power calculation derived for one test (Section 7.2) is not valid for whichever test the branch selects. A Holm correction [Holm, 1979], which controls the family-wise error rate at a chosen level a and is uniformly more powerful than Bonferroni, is applied across the specific, named comparisons of interest and not across every pairing the full $C \times C$ comparison matrix could generate, since that would over-correct for hypotheses that were never posed. Cohen's d_z, the paired-samples effect size standardised by the standard deviation of the differences themselves and equal to $t/\sqrt{n}$ for the paired t-statistic, is reported alongside every p-value; it is distinct from, and often confused with, d_{av}, which ignores the correlation between the paired measurements [Lakens, 2013]. A statistically significant but small effect is a different finding from a significant, large one.

7.2 Power and the S6 Floor

The task-count floor referenced in Section 5 is computed from the standard paired-sample size formula,

$$n^* = \frac{(z_{a'} + z_\pi)^2 \sigma^2}{\delta^2}, \tag{13}$$

one-sided, with a' the Holm-corrected per-comparison significance level (the smallest threshold the correction procedure can demand across the pre-registered comparisons; using the uncorrected family-wise a here systematically understates the requirement), z_π the standard normal quantile at the intended power π, σ the observed standard deviation of task-level paired differences from a pilot run, and δ the minimum effect size worth detecting, a decision made before the pilot runs. Statistics texts usually write the significance level as α and the power as $1 - \beta$; we use a and π because α is already the setting-cost ratio of S1 and β the batching factor of S4. Equation 13 is the one-sided form of the standard normal-approximation sample-size formula; Miller [2024] derives the two-sided version for language-model evaluations and decomposes the variance, a result this paper relies on. Writing ω^2 for the variance of the *true* task-level paired difference across tasks and σ_A^2, σ_B^2 for the within-task variance of each system's score across regenerations, the variance of an observed paired

difference averaged over R regenerations is

$$\sigma^2 = \omega^2 + \frac{\sigma_A^2 + \sigma_B^2}{R}, \tag{14}$$

which makes explicit an interaction the simplifications of Section 5 would otherwise hide: S2 (fewer regenerations) raises the second term and therefore raises n^*, so that some of what S2 saves on generation is returned as additional tasks under S6. Miller [2024] also gives the stopping rule for regenerations: once $(\sigma_A^2 + \sigma_B^2)/R \ll \omega^2$, further regenerations "will have little effect on the standard error", and the pilot's estimates of ω^2 and $\sigma_{A,B}^2$ are what determine where that point lies. Equation 13 is a starting point and no more: the test actually used (Section 7.1) is a permutation test on a finite sample, σ itself is estimated, and Bowyer et al. [2025] show that central-limit-based intervals are unreliable below a few hundred tasks. A simulation-based re-check of n^* against the permutation test, using pilot data, is therefore part of the protocol.

7.3 Blind and Calibrated Scoring

S4 (Section 5) puts two properties of the judging protocol in tension. **Blind scoring**, in which the judge is never told which system, setting or configuration produced the answer it is scoring, guards against a judge favouring a system it recognises or has been told is the "interesting" one. The concern is documented: Zheng et al. [2023] report self-enhancement bias, a judge's preference for answers produced by a model like itself, alongside verbosity bias, and Stureborg et al. [2024] report a preference for lower-perplexity, more familiar text. Blinding is compatible with either isolated or batched presentation. **Calibration context**, showing a judge several systems' answers to the same task together as in S4, addresses a different problem: a judge scoring one answer in isolation has no reference for the level of quality attainable on that specific task, which risks scale drift, where the same numerical score from two judges denotes different levels of quality because each is implicitly calibrated to a different internal reference point; Stureborg et al. [2024] document the related anchoring effect in multi-attribute judgements. Batched presentation brings its own, well-measured hazard: position bias. Wang et al. [2023a] show that the ranking of candidate answers "can be easily hacked by simply altering their order of appearance", to the point that a weaker open-weight model was preferred over a stronger proprietary assistant on 66 of 80 queries under one ordering, and Zheng et al. [2023] report the same effect; in a small measurement made for this paper, three answers graded together under an absolute rubric kept their ranking in two of three orderings, with the top two swapping places in the third, and an individual score moved by up to 17 points (Appendix C). The remedy those authors propose, balanced position calibration (scoring under several orderings and aggregating), multiplies the number of judge calls and so consumes part of what S4 saved. Batched, anonymised, order-randomised presentation under explicit absolute-scoring instructions, with the order varied across the K repeats of S3 so that the repeats double as position calibration, is our recommended way to obtain calibration context without fully sacrificing independence; the S4 discussion in Section 5 names the residual risk this does not eliminate.

7.4 Data Retention

Every individual measurement (each judge's score on each dimension, for each repeat scoring, for each regeneration) should be retained at its finest grain and never averaged down before storage. Every reported summary statistic is then a view computed from that raw table, and the raw table remains the record. This is what makes it possible to later compute a confidence interval or a p-value for any pair of systems not among the originally pre-registered comparisons, without re-running a single generation or judging call, provided that any such post-hoc comparison is reported

as *exploratory* and is not included in the same family-wise error control as the comparisons that were pre-registered before data collection. The distinction is the one Nosek et al. [2018] draw between prediction and postdiction: pre-registration "distinguishes analyses and outcomes that result from predictions from those that result from postdictions", and a retained raw table is what allows both to be reported, with the distinction intact, from the same run.

8 Empirical Validation of the Reduction Factors

Every reduction factor in Section 5 is a prediction about cost, not about the result. A simplification could save the predicted cost and change which system appears to win, or save the same amount and change nothing. Only running both versions on the same systems and tasks tells the two apart, and we recommend doing so before a simplified protocol is adopted for routine use.

Phase 0: calibration. Validate the judging protocol itself against a small set of known-good and known-poor answers, and, where the domain is high-stakes, against a small human-rated gold subset, before it is relied upon for comparison data. Agreement statistics alone do not establish this. Philipp et al. [2026] find that the best of several language-model evaluators of medical answers reached inter-rater agreement with physicians consistent with the physicians' own ceiling ($\kappa = 0.694$ against 0.709, with confidence intervals the authors call wide), yet "physicians scaled abstention with item difficulty, while frontier models assigned definitive scores in every case"; they conclude that "statistical alignment does not ensure clinical caution, and that evaluator independence requires explicit verification". The gold subset should be used to check not only whether the judge's scores agree with human raters but whether its confidence tracks item difficulty as theirs does, and the inter-judge agreement of S5 should be reported alongside.

Phase 1: full reference run. Run the complete, unsimplified protocol (E settings swept, R regenerations, J judges, K repeats, T_{full} tasks) on the systems under comparison. This run is performed once and serves as the reference against which every simplified variant is compared. It is also the pilot that the other sections refer to: it contains the two-setting, multi-regeneration and repeat-scoring subsets that S1, S2 and S3 ask for, and its data give the variance estimates from which n^* is set (Section 7.2) and the setting-cost ratio α on the benchmark's own tasks (Appendix C).

Phase 2: each intermediate simplification, on the same systems and tasks. Run S1 alone, S2 alone, and so on, and the fully combined simplification from Section 6, each against the identical task set used in Phase 1. For each variant, record: whether the same system(s) rank highest; how far the point estimate and confidence interval shift relative to the Phase 1 reference; and whether the corrected p-value for the pre-registered comparisons crosses the significance threshold in the reference run but not in the simplified one, or vice versa. Rank agreement between a reduced and a full protocol is the criterion the benchmark-compression literature uses to certify a subset: Saranathan et al. [2025] select subsets of 1–20% of a benchmark that preserve full-benchmark rankings, reporting evaluation-cost reductions of 80–99%, and Perlitz et al. [2024a] score each design choice by its effect on ranking reliability. Phase 2 should report the same statistic, Kendall's τ over the systems under comparison, alongside the per-comparison outcomes. Comparing two protocols on the same systems is formally a benchmark-agreement test, for which Perlitz et al. [2024b] show that "overlooked methodological choices can significantly influence" the result, in particular which systems are included and which protocol is treated as the reference, and propose standard practices. The full Phase 1 run is the reference here by construction, and the systems compared should be the full set under study and not a convenient subset.

Phase 3: fidelity cost per simplification, reported alongside cost saved. The useful output of this protocol is a table of cost saved against fidelity lost, per simplification, from which a future user of this framework can select the combination that matches their tolerance for each. That combination may reasonably differ from the one in Section 6; which trade-offs are acceptable is a decision the paper's formulas inform without making.

9 Limitations and Open Questions

The generation-cost parameter g is treated as a single external number per system per setting; in practice it may itself have meaningful variance (a system's answer length, and therefore its cost, can differ substantially task to task), which the framework absorbs into the variance term σ used for power calculations instead of modelling it separately; a more complete treatment would model the distribution of g. The batching saving in Equation 8 assumes a judge can meaningfully score C answers in one call without a quality cost from context length or attention dilution as C grows large; this should be checked empirically for large C. The concern has independent support: Hsieh et al. [2024] find that "almost all models exhibit large performance drops as the context length increases", with only half of the models they test maintaining satisfactory performance at 32K tokens, and Liu et al. [2024a] document the position-dependent degradation to which a multi-answer prompt is exposed. The nested measurement structure (R regenerations $\times$ J judges $\times$ K repeats, aggregated to one score per system per task) removes real variance at each aggregation step before the final variance calculation in Equation 13. In the language of generalisability theory [Brennan, 2001], tasks, judges, judge repeats and regenerations are *facets* of the measurement, each contributing a variance component that a generalisability study estimates from a fully crossed pilot; the paired-difference interval of Section 7 generalises over the task facet only and is conditional on the particular judges, presentation orders and decoding configuration used. That is the sense of the argument in S5: at $J = 1$ the judge facet's variance component is not zero but unidentifiable, and the same holds for the regeneration facet at $R = 1$ and the repeat facet at $K = 1$, which is why Section 5 asks for each to be estimated on a pilot subset before it is dropped. A mixed-effects model with crossed random effects for task and judge [Baayen et al., 2008], with regeneration and repeat nested within them, would propagate all components into a single interval, at the price of simplicity; the interval reported here is conditioned on the specific judge panel and aggregation rule. The raw table retained under Section 7 is what makes fitting such a model later possible without re-running anything. The minimum effect size δ worth detecting for a given comparison remains a domain-specific judgement, which the formulas in Section 7.2 take as an input.

10 Conclusion

Rigour in a DRACO-style benchmark is expensive by construction, because its cost is a product of six design choices, and most of that product can be given back. The sweep over settings, the regenerations and the task count carry most of the saving. Batched judging carries far less than its unbatched arithmetic suggests once prompt caching is priced in. The judge count, the dimension intuition reaches for first, is the one to keep. The combined reduction is a weighted harmonic mean of a generation-side and a judging-side factor, roughly two orders of magnitude in the worked instance of Appendix B and less at the setting-cost ratios we measured. What that buys is a benchmark cheap enough to re-run whenever the set of systems changes, provided the reduced protocol is checked against the full one at least once (Section 8).

A Published DRACO-Style Runs

This appendix records the named, dated figures behind Sections 2 and 3. Each was checked against the primary page cited; where a primary page no longer resolves, the entry records this. All access dates are September 2026, except that the Microsoft post was re-checked at its new address on 25 October 2026.

A.1 The benchmark

DRACO was published by Zhong et al. [2026] (Perplexity AI; arXiv, 12 February 2026), with the task set released under an MIT licence [Perplexity AI, 2026a]. The task pipeline began from 1,000 high-difficulty English queries drawn from the company's own deep-research product and was filtered to 100 tasks that the authors describe as "anonymized, open-ended and complex, objectively evaluable, and representative", spanning 10 domains and drawing on information sources from 40 countries. Each task carries a rubric averaging roughly 40 binary criteria across four dimensions. The paper's primary judge was Gemini 3 Pro, with GPT-5.2 and Claude Sonnet 4.5 as secondary judges; each answer was graded in five independent runs and the standard deviation across runs is reported. The authors' Appendix A notes "systematic differences in absolute score levels" between judges (GPT-5.2 "consistently assigned lower scores"), and its Table 15 gives each of the seven systems' scores under all three judges: GPT-5.2 scored them 14.8 to 23.9 points below Gemini 3 Pro and 16.4 to 27.4 points below Sonnet 4.5, while those two judges differed by 0.2 to 12.0 points. The paper's Table 8 reports, for example, the company's own Deep Research product (backed by Claude Opus 4.6) at $70.5 \pm 0.3\%$; the company's blog post on the benchmark [Perplexity AI, 2026b] could not be retrieved by automated means at the time of writing; the benchmark paper's own tables are therefore cited for every figure, and the post is cited for completeness only.

A.2 Published scores

The Microsoft replication [Anand, 2026] is a Microsoft Tech Community post of 30 March 2026 by Gaurav Anand; its statistics are independently quoted by Thomas [2026], which also reports the per-dimension gains (+3.33 breadth and depth, +3.04 presentation, +2.58 factual accuracy). It is the only published DRACO-style run we found that reports a standard error, a significance test, and the number of judge repeats together. The OpenRouter post [OpenRouter, 2026] of 12 June 2026 reports no error bars; it states that its judge was Gemini 3.1 Pro Preview "instead of the paper's choice of Gemini 3 Pro", that the budget panel was synthesised by the same model as the main ensemble, and that "the paper reports 10–25 point shifts between judges". That range is not stated in the DRACO paper, but it roughly summarises the paper's Table 15, where GPT-5.2 scores each system 14.8 to 27.4 points below the other two judges; the range should therefore be read as the aggregator's summary of the authors' own figures, not as a separate measurement. The Valyu post [Valyu, 2026] describes multi-step retrieval across specialised data sources and does not describe multi-model ensembling, states neither its judge model nor its run date, and reports no error bars, but does state that its datasets, run scripts, raw outputs and judge evaluations are published, which is the retention practice Section 7 requires; it is the only one of the four sources to state this.

A.3 Third-party trackers

Two public leaderboards that aggregate DRACO results were consulted on 10 and again on 11 September 2026. One [LLM Stats, 2026] showed a top score of 0.772 "across 3 evaluated AI models", all three self-reported and none verified; the other [BenchLM, 2026] showed 88.6% for the leading

System	Type	Score	Judge	Precision reported
Valyu DeepResearch, "Heavy" mode [Valyu, 2026]	retrieval-centred product	72.7%	not stated	none; date not stated
Perplexity Deep Research (Opus 4.6) [Zhong et al., 2026]	benchmark author's product	70.5%	Gemini 3 Pro	±0.3 SD over five runs
OpenRouter "Fusion" (Fable 5 + GPT-5.5, synthesised by Opus 4.8) [OpenRouter, 2026]	synthesis ensemble	69.0%	Gemini 3.1 Pro Preview	none; one constituent completed 93 of 100 tasks (content filter)
OpenRouter budget panel (three inexpensive models, synthesised by a fourth) [OpenRouter, 2026]	synthesis ensemble	64.7%	Gemini 3.1 Pro Preview	none
Microsoft 365 Copilot Researcher "Critique" [Anand, 2026, Thomas, 2026]	multi-model research assistant	57.4	GPT-5.2 ("the strictest of the three")	+7.0 (SEM ±1.90) over Perplexity Deep Research under the same judge; against its own single-model version, paired t-test $p < 0.0001$ on every dimension and significant in 8 of 10 domains; five runs per question

Table 1: Published DRACO scores as of September 2026, in descending order. Scores under different judges are not comparable: the Microsoft figure was produced by the judge the benchmark's own authors identify as the strictest, the OpenRouter figures by a newer model than the paper's primary judge. The single-model scores in the OpenRouter post (Fable 5 65.3, DeepSeek V4 Pro 60.3, GPT-5.5 60.0, Opus 4.8 58.8) are what the ensembles are compared against.

entry among three submitted system set-ups. Neither figure appears in any of the primary sources above, and the two trackers disagree with each other about which system leads. The first page showed 0.732 "across 1 evaluated AI model" when first consulted and 0.772 "across 3" a day later, which illustrates the point made in Section 3.

B A Worked Cost Instance at Current Commercial Terms

This appendix instantiates the algebra of Sections 4–7.2 with concrete, dated, sourced values.

B.1 Prompt-caching terms

At the time of writing (September 2026) the three largest commercial API providers bill a repeated prompt prefix as follows. OpenAI charges cached input tokens at 0.1× the standard input price, applies caching automatically to prefixes of at least 1,024 tokens, and keeps a prefix eligible for reuse for up to 30 minutes [OpenAI, 2026a]. Anthropic charges cached reads at 0.1× (0.025× for its newest models), bills the first, cache-writing call at 1.25× for a five-minute cache lifetime, and requires minimum prefixes of 512–4,096 tokens depending on model [Anthropic, 2026a]. Google charges

cached reads at 0.1× and adds a per-hour storage charge for explicitly created caches [Google, 2026]. None of the three caches output tokens. In the notation of Equation 5, the representative values are $c = 0.1$ and $w = 1$ (OpenAI, Google) or $w = 1.25$ (Anthropic). All three pages were re-read on 11 September 2026: OpenAI's terms apply to its current model generation, Anthropic bills a one-hour cache write at 2×, and Google's storage charge ranges from $0.50 to $1.80 per million tokens per hour depending on model tier.

B.2 The setting-cost ratio α

Artificial Analysis ran a fixed evaluation suite against one frontier model, GPT-5, at each of its four reasoning-effort settings. The suite consumed 3.5 million output tokens at the `minimal` setting and 82 million at `high`, a 23× difference, for Intelligence Index scores of 44 and 68 respectively [Artificial Analysis, 2025]. Because reasoning tokens are billed as output tokens at the same rate regardless of setting [OpenAI, 2026b, Anthropic, 2026c], this is a measured $\alpha \approx 23$ for that model on that suite. The same service publishes per-task token usage for each model it evaluates [Artificial Analysis, 2026]; α should be looked up or measured for the systems under comparison. Appendix C reports such a measurement on analytic prompts, with α between 1.75 and 3.2, and its consequence for the worked instance below.

B.3 Call counts for a modest configuration

For $C = 5$ systems, $T = 60$ tasks, $E = 2$ settings, $R = 3$ regenerations, $J = 3$ judges and $K = 5$ repeats, Equation 1 implies $C \cdot E \cdot R \cdot T = 1{,}800$ generation calls, Equation 2 implies 27,000 unbatched judging calls, and Equation 3 implies 5,400 batched judging calls, before any per-call price is attached.

B.4 The batching factor β

With $C = 5$ and $\rho = 0.2$, Equation 8 gives $\beta = 3.0$ without caching ($c = w = 1$). With $c = 0.1$ and $w = 1$ it gives $\beta = 1.2$, and the ceiling $1 + (C - 1)\,c$ is 1.4; with $w = 1.25$, $\beta = 1.325$. Batching therefore saves a factor of three on judging cost without caching and, with it, a factor of 1.2, about a sixth of the judging cost.

B.5 The combined reduction

Take $\alpha = 20$, $R = 3$, $K = 5$, $E = 2$, $C = 5$, $\rho = 0.2$, and a full task count $T_{\text{full}} = 180$ reduced to an assumed floor $n^* = 60$. Then Equation 11 gives $F_G = 21{\cdot}3{\cdot}3 = 189$ and $F_J = 2{\cdot}3{\cdot}5{\cdot}3{\cdot}\beta = 90\beta$, that is 270 with uncached batching and 108 under cached-read pricing. Table 2 evaluates Equation 12 across plausible values of the judging share s. The floor of 60 is assumed for illustration and does not follow from the power calculation below, whose inputs give $n^* = 128$ at $R = 3$ and $n^* = 184$ once S2 reduces R to 1, as it does here. On those inputs the floor exceeds $T_{\text{full}} = 180$, S6 saves nothing (T_{full}/n^* is at most 1, not 3), and because F_G and F_J both carry that factor, every reduction in the table falls to a third, 46–74×.

B.6 The task floor n^*

For three pre-registered comparisons at family-wise $a = 0.05$, Holm's smallest per-comparison threshold is $a' = 0.05/3$; one-sided, $z_{a'} \approx 2.13$. At power $\pi = 0.8$, $z_\pi \approx 0.84$. With a pilot

Judging share of full cost, s	0.1	0.3	0.5
Reduction, no prompt caching ($F_J = 270$)	195×	208×	222×
Reduction, cached-read pricing ($F_J = 108$)	176×	154×	137×

Table 2: Combined reduction from S1–S4 and S6 (Equation 12) for the worked instance, with $F_G = 189$. Multiplying every factor against the total cost would give $(1+\alpha)\cdot E\cdot R\cdot K\cdot\beta\cdot T_{\text{full}}/n^* = 5{,}670\times$ for the same inputs without caching and 2,268× with it, the overstatement Section 6 describes. The reference implementation reproduces every entry of this table (`tests/simplifications.test.mjs`).

standard deviation of task-level paired differences $\sigma = 19$ points on a 100-point scale and a minimum effect worth detecting of $\delta = 5$ points, Equation 13 gives $n^* = 128$ tasks. By Equation 14, if that σ was measured at $R = 3$ with within-task variances $\sigma_A^2 = \sigma_B^2 = 120$ and between-task variance $\omega^2 = 281$, then dropping to $R = 1$ (S2) raises σ^2 from 361 to 521 and n^* from 128 to 184, the interaction Section 7.2 describes. The reference implementation reproduces both figures (`tests/power.test.mjs`).

C Measurements Made for This Paper

The appendices above rely on published figures. Three premises of the cost model are inexpensive enough to be measured directly, and were measured on 11 September 2026 with a small, pre-stated budget; the scripts, prompts and raw provider usage records are in the repository under `experiments/`; `experiments/MANIFEST.pdf` lists every run with its purpose, scripts, models, stated and actual spend, result files and their SHA-256 hashes, and caveats. Every figure below is a single small measurement made on one day and is reported so that a reader can judge the premise; the validation protocol of Section 8 is where such measurements belong at scale.

C.1 Judging-side premises (experiment E2)

Set-up. The first of the five E1 prompts (listed in Section C.3), a rubric of 40 binary criteria in the DRACO style, three short candidate answers of deliberately different quality, and a frontier model as judge at its lowest reasoning setting, at the provider's default sampling (the model rejects a temperature parameter). Total spend: $0.19 across eleven calls.

Prompt caching. Three consecutive calls shared the rubric prefix and differed only in the answer. The provider reported no cached tokens on any of the three: the shared prefix was roughly 920 tokens, below the provider's documented 1,024-token minimum (Appendix B). Five later calls with an identical full prompt of 1,092 tokens each reported 1,089 of them as cached. Equation 5 therefore holds only once the fixed prefix clears the provider's minimum; a rubric of forty short criteria does not do so on its own, and a protocol relying on the cached rate should measure its prefix length. The same eleven calls give the fixed and marginal token counts of Equation 8: about 920 fixed tokens and about 130 marginal tokens per additional short answer, $\rho \approx 0.14$; answers of the length deep-research tasks produce would raise ρ considerably.

Test–retest (S3). The same answer was scored five times with no memory between calls. The scores were 50, 50, 60, 50 and 43 on the 100-point scale: mean 50.6, standard deviation 6.1, range 17 points. A single scoring of this answer could have fallen anywhere within a 17-point interval, which is the quantity S3 says is lost at $K = 1$.

Order sensitivity (S4, Section 7.3). The three answers were scored together in one call, under explicit instructions to grade each independently against the absolute rubric, in three rotations of their order. The ranking of the three answers was the same in the first two orderings; in the third, the strongest answer scored two points below the middle one, and the weakest stayed last. The strongest answer scored 60, 65 and 48 depending on position, the middle one 45, 55 and 50, the weakest 25, 25 and 20. Order alone moved an absolute score by up to 17 points and, in one ordering, swapped the top two answers. Section 2.2 notes that the choice of judge moves published DRACO scores; here the order of presentation moved them within one judge, and that is the reason Section 7.3 asks for the order to be varied across the K repeats.

C.2 The setting-cost ratio α (experiment E1)

Set-up. Five open-ended analytic prompts of a business-analysis kind (banking, mining, automotive, retail and telecommunications scenarios, each asking for a method and its most likely failure mode), one system instruction asking for a thorough professional answer, and two frontier models, one from each of the two largest providers, each queried once per prompt at each of its documented reasoning-effort settings: `none`, `low` and `high` for the first, `low` and `high` for the second. Output was capped at 24,000 tokens; no call reached the cap. Six calls failed with transient provider errors and

were retried until they succeeded, so that every cell is filled. Total spend: $2.17 across 25 successful calls.

Result. Table 3 gives the paired ratios. For the first model the `high` setting consumed 2.8× the output tokens of `low` (per prompt 1.5–4.0) and 3.2× those of `none`; at `high` the provider reported between 1,024 and 4,068 reasoning tokens per answer, at `none` zero, and `low` added only 12% over `none`. For the second model `high` consumed 1.75× the tokens of `low` (per prompt 1.5–2.0), with mean answers of 3,600 and 6,300 tokens. Cost ratios are slightly lower than the token ratios (2.80, 3.14 and 1.74 against 2.83, 3.18 and 1.75; both are 1.12 for `low` against `none`), because the same input cost is added at both settings. Mean latency rose from 38 seconds at `none` to 125 at `high` for the first model and from 58 to 97 seconds between `low` and `high` for the second.

Model	Settings compared	Paired prompts	α by tokens	Per-prompt range
first provider	high / low	5	2.83	1.52–3.99
first provider	high / none	5	3.18	2.04–4.25
first provider	low / none	5	1.12	0.94–1.34
second provider	high / low	5	1.75	1.48–2.00

Table 3: Measured setting-cost ratios on five analytic prompts, 11 September 2026. Pooled ratio of summed output tokens over the same five prompts; raw usage records in `experiments/results/`.

Interpretation. The published 23× of Appendix B was measured on a reasoning-benchmark suite; on analytic prose tasks the same two providers' effort settings change cost by a factor of two to three. α is therefore not a property of a model but of a model on a task family, and the pilot of Section 8 should measure it on the benchmark's own tasks before S1's saving is counted. The consequence for the worked instance of Appendix B is material: replacing its illustrative $\alpha = 20$ with the measured values gives a generation-side factor F_G of 25 to 38 instead of 189, and a combined reduction (Equation 12) of roughly 27–66× across the same range of s and caching assumptions instead of 137–222×, about one and a half orders of magnitude instead of two. If the expensive setting is the one retained (Section 5, S1), the same configuration yields 13–27×. The framework's conclusion that the saving is dominated by the generation side and that judge count is the wrong place to cut is unchanged; the size of the overall saving is set by α, and α must be measured.

C.3 Full method for experiment E1

This subsection gives the information required to reproduce E1 independently of the repository; the scripts (`experiments/e1-alpha.mjs`, `e1-retry.mjs`), the prompt file and the raw provider responses' usage blocks are also committed.

Models and interfaces. First provider: OpenAI, model `gpt-5.6-sol`, endpoint `POST https://api.openai.com/v1/chat/completions`, request fields `model`, `messages` (one `system` and one `user` message), `max_completion_tokens` = 24000, `reasoning_effort` ∈ {`none`, `low`, `high`}; no temperature field (the model rejects it). Second provider: Anthropic, model `claude-opus-5`, endpoint `POST https://api.anthropic.com/v1/messages`, header `anthropic-version` = `2023-06-01`, fields `model`, `max_tokens` = 24000, `system`, `messages` (one `user` message), `output_config.effort` ∈ {`low`, `high`}; no thinking budget set explicitly and no temperature field. Model identifiers and prices ($5 input and $30 output per million tokens for the first; $5 and $25 for the second) are the providers' list prices on 11 September 2026.

System instruction (verbatim). "You are a careful analyst. Answer the question thoroughly for a professional reader: state your method, the key assumptions, the most likely failure modes, and what evidence would change your conclusion. Use headings and be concrete."

User message. The prompt's context, a blank line, then "Question: " followed by the question. The five prompts, verbatim:

1. **banking** We want to know whether a collections strategy introduced in March reduced 90-day default rates. Accounts were assigned by branch, not randomly, and the branches that adopted early had the worst arrears to begin with. We have 30 months of monthly account-level data. *Question:* What method would you use, and what is the most likely way this analysis produces a confidently wrong answer?
2. **mining** We want to know whether a new blast design improved fragmentation. It was trialled on 30 of 200 blasts, chosen by the drill and blast engineers in areas they judged suitable. Fragmentation is measured by image analysis on the muckpile, which is affected by how the shovel digs. *Question:* How would you estimate the effect, and what is the most likely way this analysis misleads?
3. **automotive** Warranty claims for a component are rising. Claims are recorded at dealer level with free-text fault descriptions, build dates come from a separate system, and a supplier change happened partway through the model year. Vehicles built later have had less time in service. *Question:* How would you establish whether the supplier change caused it, and what is the most likely false conclusion?
4. **retail** We have 24 months of weekly units sold by store and SKU, about 6 million rows. A price cut was applied to one category in week 60, but only in stores above a footfall threshold. Some stores closed for refurbishment during the period. *Question:* What method would you use, and what is the most likely way this analysis produces a confidently wrong answer?
5. **telecoms** We want to know whether a network upgrade reduced churn. It rolled out cell by cell over a year, prioritising the worst-performing cells. Customers are assigned to a home cell by billing address, which is often not where they actually use the service. *Question:* How would you estimate the effect, and what is the most likely way this analysis misleads?

Procedure. Calls were made sequentially in the order model × effort × prompt, one call per cell, on 11 September 2026 between 07:41 and 08:20 UTC (the first run) and 08:55 to 09:10 UTC (retries). Six calls failed with the provider's HTTP 500 "server error" and were retried, up to three times with a delay, until they succeeded; the retried cells are marked in Table 4. No call was excluded. No call reached the output cap (all finish reasons `stop` or `end_turn`).

Quantities counted. Output tokens are the provider's own `usage` fields: `completion_tokens` (first provider; this includes the reasoning tokens, which the provider also reports separately as `completion_tokens_details.reasoning_tokens`) and `output_tokens` (second provider; reasoning is not itemised). Input tokens are `prompt_tokens` and `input_tokens`. Cost is input tokens times the input price plus output tokens times the output price, at the prices above; no cached-input discount applied (each prompt was sent once per setting, and the shared system prefix is below the caching minimum). Latency is wall-clock time of the HTTP request.

Analysis. For each model and pair of settings, α is the ratio of the *sums* of output tokens over the five prompts (pooled ratio); the per-prompt ratios are reported alongside as the range. The cost ratio is computed the same way; it is slightly lower than the token ratio, by at most 0.04, because the identical input cost is added to both sums. No statistical test is applied: with five prompts the result is a point estimate with a stated range, offered as evidence that α is task-dependent and not

as an estimate of α for any task population.

Comparison with the published 23×. The published figure [Artificial Analysis, 2025] is not from the DRACO paper, which does not measure α; it is the total output-token consumption of running an evaluation suite. The two measurements differ in every design dimension, so their disagreement is not a contradiction but an open question, taken up in Section C.4:

	Published (Artificial Analysis)	This paper (E1)
Model	GPT-5 (2025)	gpt-5.6-sol; claude-opus-5 (2026)
Settings	`minimal` vs `high`	`none`/`low` vs `high`; `low` vs `high`
Tasks	the Intelligence Index suite (reasoning, knowledge and coding benchmarks; composition per their methodology page)	five open-ended analytic prose prompts
Tools / browsing	per suite	none
Prompts	thousands (suite total)	five
Repeats	per suite	one call per cell
Token accounting	suite-total output tokens	provider usage fields per call
Instruction	benchmark harness prompts	"answer thoroughly, use headings"
Result	23× (3.5M → 82M)	2.83× (3.18× vs `none`); 1.75×

C.4 Why the two measurements of α differ

Four candidate explanations for the distance between the published 23× and the measured 1.75–3.2× are set out below; none of them is a contradiction, and together they make α a quantity that must be measured on the tasks of interest. Experiment E3 (Section C.5) tests the first of them and does not support it.

Task family. The published figure was measured on an evaluation suite composed of "MMLU-Pro, GPQA Diamond, Humanity's Last Exam, LiveCodeBench, SciCode, AIME, IFBench & AA-LCR" [Artificial Analysis, 2025], most of which are competition-, examination- or code-style problems with a short, checkable answer. On such problems nearly all of the output at a high setting is reasoning, and the literature on reasoning length shows the amount of reasoning a model spends varies by problem type and across model generations: Ballon et al. [2026] find that "new generations of reasoning models use test-time compute more effectively" and that accuracy declines as reasoning chains grow even at fixed difficulty; Lee et al. [2025] show that "there exists an optimal amount of reasoning tokens such that the probability of producing a correct solution is maximized" and that it grows with the length of the underlying computation a task requires; and Du et al. [2025] report that "models solving the same problem with similar accuracy can exhibit up to a 5.0× difference in token length". Our prompts asked for a structured analysis in prose, where the visible answer is itself thousands of tokens at every setting.

Arithmetic of the ratio. Because α is a ratio of *total* output tokens, a large visible answer compresses it. In E1 the first model produced about 1,150 tokens at `none` with zero reasoning tokens and about 3,650 at `high` of which 1,024–4,068 were reasoning: measured on reasoning tokens alone the ratio is unbounded, measured on total tokens it is 3.2. A benchmark whose answers are a single number has almost no visible floor, so the same reasoning behaviour yields a far larger total-token ratio. This follows from the definition and says nothing about the models.

Effort semantics. Both providers now document adaptive behaviour that narrows the gap between settings on tasks that do not need long reasoning. OpenAI's guide states that its models "reason

Model	Effort	Prompt	In	Out	Reasoning	Latency (s)	Stop
claude-opus-5	high	1 banking	191	5704	–	89	end_turn
claude-opus-5	high	2 mining	199	6641	–	106	end_turn
claude-opus-5	high	3 automotive	184	6702	–	102	end_turn
claude-opus-5	high	4 retail	183	6071	–	91	end_turn
claude-opus-5	high	5 telecoms	180	6298	–	97	end_turn
claude-opus-5	low	1 banking	191	3370	–	54	end_turn
claude-opus-5	low	2 mining	199	3543	–	57	end_turn
claude-opus-5	low	3 automotive	184	3358	–	55	end_turn
claude-opus-5	low	4 retail	183	3413	–	54	end_turn
claude-opus-5	low	5 telecoms	180	4262	–	68	end_turn
gpt-5.6-sol	high	1 banking	129	2464	1536	138	stop (retry 1)
gpt-5.6-sol	high	2 mining	133	4062	3072	126	stop (retry 3)
gpt-5.6-sol	high	3 automotive	125	2251	1024	73	stop
gpt-5.6-sol	high	4 retail	128	5203	4068	160	stop
gpt-5.6-sol	high	5 telecoms	125	4306	2920	130	stop
gpt-5.6-sol	low	1 banking	129	1343	341	43	stop
gpt-5.6-sol	low	2 mining	133	1166	139	41	stop (retry 1)
gpt-5.6-sol	low	3 automotive	125	1485	352	48	stop
gpt-5.6-sol	low	4 retail	128	1303	325	43	stop
gpt-5.6-sol	low	5 telecoms	125	1164	100	41	stop (retry 1)
gpt-5.6-sol	none	1 banking	129	1105	0	39	stop (retry 2)
gpt-5.6-sol	none	2 mining	133	1080	0	39	stop (retry 2)
gpt-5.6-sol	none	3 automotive	125	1105	0	34	stop
gpt-5.6-sol	none	4 retail	128	1225	0	37	stop
gpt-5.6-sol	none	5 telecoms	125	1234	0	39	stop

Table 4: Every call of experiment E1 (25 calls, 11 September 2026). In/Out/Reasoning are the providers' own usage fields; "Reasoning" is included in "Out" for the first provider and not reported by the second. Stop = provider finish reason; no call stopped on the output cap.

adaptively across reasoning efforts, using fewer tokens for simpler tasks and thinking harder for complex tasks" [OpenAI, 2026b]. Anthropic's states that "effort is a behavioral signal, not a strict token budget", that "at lower effort levels, Claude still thinks on sufficiently difficult problems, but thinks less", and, for the model measured here, that "effort controls thinking volume, not visible response length: on Claude Opus 5, changing effort does not reliably shorten responses" [Anthropic, 2026b]. A setting that behaves this way cannot produce a 23× spread on prose tasks; the measured 1.75× is what the documentation predicts.

Model generation and the published suite itself. The 23× was measured on the first model of its series in 2025; the models measured here are a later generation, and the provider that published the suite has since changed the suite's composition, weighting and evaluation mechanics [Artificial Analysis, 2026], so the published number is not one that its own publisher would reproduce today on the same terms.

What would discriminate these. Two cheap measurements: the five E1 prompts on the 2025 model at `minimal` and `high`, which isolates model generation; and a set of competition-style problems through the E1 models at `low` and `high`, which isolates task family. The second is reported below as experiment E3.

C.5 Competition problems as a task-family test (experiment E3)

Set-up. The same two models and the `low` and `high` settings of E1, applied to the first five problems of the 2025 American Invitational Mathematics Examination (paper I), taken from the public dataset `math-ai/aime25` through the Hugging Face datasets API; system instruction: "Solve the competition mathematics problem. Show your reasoning briefly, then give the final answer as an integer on the last line in the form 'ANSWER: <integer>'." Output cap 16,000 tokens, never reached. One call per cell; six calls, all to the first provider, failed with transient HTTP 500 errors during an outage (three attempts each) and were retried until they succeeded. Total spend: $0.39 across 20 calls. The scripts (`e3-aime.mjs`, `e3-retry.mjs`), the problems and the raw usage records are committed alongside E1's.

Result. If task family explained the gap between the published 23× and E1, α should be far larger here than on prose. It is not (Table 5). For the first model the pooled ratio is 1.48 (median per problem 1.18; reasoning tokens 105–1,435 at `high` against 114–526 at `low`), *lower* than its 2.83 on prose; all ten answers were correct. For the second model the pooled ratio is 2.67 but heavy-tailed: the per-problem ratios are 0.97, 1.22, 1.76, 2.12 and 72.7, the last from a problem answered correctly in 26 tokens at `low` and in 1,890 at `high`; the median is 1.76, close to its pooled prose value of 1.75; accuracy was four of five at `low` and five of five at `high`.

Model	Setting	Out tokens	Pooled α	Median α	Correct	Per-problem α
gpt-5.6-sol	low	2,675			5/5	
gpt-5.6-sol	high	3,959	1.48	1.18	5/5	1.01, 2.02, 1.18, 1.11, 1.33
claude-opus-5	low	1,857			4/5	
claude-opus-5	high	4,960	2.67	1.76	5/5	1.76, 72.7, 0.97, 1.22, 2.12

Table 5: Experiment E3, 11 September 2026: the E1 models and settings on AIME 2025 I problems 1–5. Pooled α is the ratio of summed output tokens; per-problem values are in problem order.

Interpretation. E3 does not support task family as the explanation for these models on these items, and it exposes a confound that any such test must reckon with: a competition problem that

a frontier model answers correctly in 26 tokens is not, for that model, a hard problem, and a public 2025 problem set is likely to have been seen in the training of models released in 2026. What E3 does show is consistent with the effort-semantics and model-generation explanations of Section C.4: for these models the `low` setting already spends reasoning where the model judges it necessary and the `high` setting adds little where it does not, so the pooled ratio stays between about one and a half and three on both task families (1.48 to 2.83), with a heavy tail on individual items. The published 23× is therefore best read as a property of a particular 2025 model, whose effort levels behaved as coarse switches, on a suite that was hard for it at the time, and not a constant of the effort setting. A test of task family that survives this confound would need problems demonstrably unseen by the models and hard enough that `low` fails on them, which the pilot of Section 8 can supply from its own retained data.

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